



Embedded Systems Testing To Ensure Successful Products

The primary motive of this article is to sensitise both hardware and software designers about the importance of designing testable products and how to enable low cost of testing

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In my more than three decades of experience, especially the last five years of working with startups and small and medium enterprises (SMEs), I have found that one of the important issues that most of them face is insufficient testing of embedded system products. They pay less attention to comprehensive testing of the products they develop, especially when the product has hardware in a fairly large amount.

An often-repeated statement is that it is impossible to test a product hundred percent. While this may be true to some extent for software in an embedded system, the hardware component has to be tested hundred percent to ensure the products are reliable and work as designed.

A second fallacy is to plan the testing of the systems after the design is complete; this never guarantees a fully tested product.

The third most common issue that I have noticed in my career is the complete ignorance of precision and accuracy of measuring circuitry. Almost all designers ignore that the

system needs to be designed for precision and accuracy, and that the provision for calibration has to be in the design and not from provided components.

Many a time I have seen designers telling me that a certain issue be addressed by the manufacturing team. Manufacturing teams can only produce what you have given 'as is' and cannot improve the design.

This article highlights the need for understanding of testing of embedded systems at various stages. It covers the different types of tests that need to be done at various stages from development to manufacturing, and their importance. The idea is to help designers appreciate the value of the testing of embedded systems.

Anatomy of an embedded system

To understand the testing of embedded systems, we need to understand them at a high level. Refer Fig. 1 which shows the high-level functional blocks of a typical embedded system. Notice the testing of the system's power supply and battery backup (#6 & #7)—most modern systems are battery backed-up systems and have become critical in the recent times. Remember that for a successful embedded system, every functional block has to be tested fully before it is shipped to the market. A typical embedded system has four major parts:

1. Core of the embedded system is the electronics which can be a single printed circuit board assembly (PCBA) or multiple PCBAs.

2. Mechatronics components like relays, motors, solenoids, etc.

3. Displays and keyboards (typically

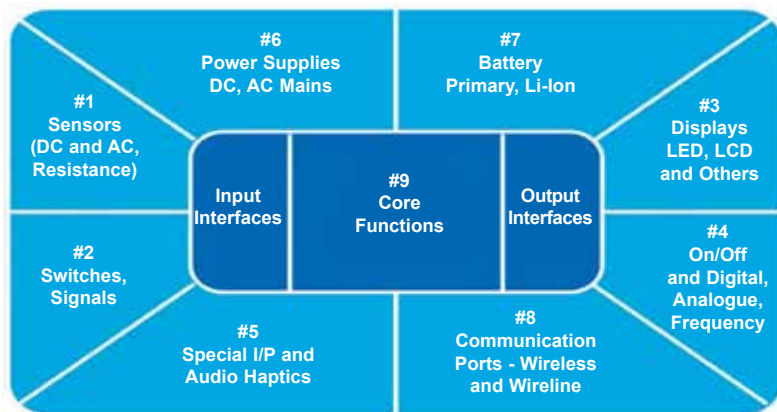


Fig. 1: High-level functional blocks of a typical embedded system